

NIKHIL GUPTA

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nikhilguptaml.vercel.app

TECHNICAL SKILLS

Languages: Python, C, C++

AI/ML Frameworks: LangChain, LangGraph, FAISS, Ollama, Streamlit, FastAPI, SQLAlchemy, ChromaDB

LLM/VLM Stack: LoRA, QLoRA, Unsloth, HuggingFace Transformers, llama.cpp, Qwen models, Groq SDK

Tools: Git, GitHub, MySQL

PROJECTS

Document Virtualizer | VLM Finetuning Pipeline | Internship Project

Jan 2026 – Present

- Designed an end-to-end QLoRA finetuning pipeline for Qwen2.5-VL targeting $\geq 95\%$ field extraction accuracy on structured government forms and handwritten content.
- Built annotation pipelines, data versioning workflows, and GGUF export for llama.cpp deployment.

Cognitive Load Balancer (CLB) | FastAPI · ChromaDB · MediaPipe

2026

- Built a real time adaptive learning system that detects cognitive state from keystroke dynamics, MediaPipe facial metrics, and response latency to dynamically adjust quiz difficulty.
- Architected REST API backend on FastAPI with SQLAlchemy and ChromaDB; debugged and resolved document upload endpoint failures under concurrent load.

EXPERIENCE

AI Engineer [Dr . Shiv Ram Meena], [USICT, Delhi]

Jan 2026 – Present

- Designed and implemented an end to end finetuning pipeline for government document understanding using Qwen VLM models, targeting $\geq 95\%$ field extraction accuracy on structured forms and handwritten content.
- Applied LoRA and QLoRA frameworks , freezing MLP modules while finetuning vision layers and attention heads to optimize GPU memory utilization.
- Coordinated with a university professor and a government stakeholder; established zero-shot evaluation baselines, data versioning, and annotation pipelines for ~ 200 image diagnostic dataset.

EDUCATION

B.Tech in Computer Science — [Vivekananda Institute of Professional Studies , Delhi]

2024-2028

NPTEL Certifications (IIT-affiliated, proctored): *Introduction to Machine Learning(Elite)* · *Introduction to LLMs(Elite)* · *NLP and Computer Vision (Elite + Gold)*

ACHIEVEMENTS & CERTIFICATIONS

- Completed Andrew Ng's Machine Learning Specialization and Deep Learning Specialization (Coursera / DeepLearning.AI).
- Participated in Smart India Hackathon (SIH) — national level government sponsored hackathon.